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Monetary Policy Transmission in Segmented Markets

Contents

1 Big picture

- **Question.** How efficiently do central bank policy rate changes pass through to repo rates when repo markets are segmented and OTC customers must trade through dealer banks?
- **Market.** The paper studies the euro-area repo market using the ECB Money Market Statistical Reporting dataset, which contains transaction-level trades by large reporting dealers.
- **Segmentation.** Dealer banks have access to centralized inter-dealer trading platforms, while many non-dealer banks and non-bank customers trade over the counter with only a small number of dealers.
- **Market power.** Because OTC customers form few dealer links and relationship formation is costly, dealers have bargaining power over OTC repo customers.
- **Main empirical fact.** Identical or nearly identical repo contracts backed by the same collateral trade at substantially different rates across customers.
- **Net interest margin fact.** Dealers lend to OTC customers at systematically higher rates and borrow from OTC customers at systematically lower rates than comparable inter-dealer rates.
- **Passthrough fact.** Following the ECB's September 2019 deposit facility rate cut, dealers passed through only about 79.4% of inter-dealer rate changes to repo depositors and about 71.7% to repo borrowers.
- **Inequality fact.** Passthrough is better for customers with more dealer relationships. Customers connected to fewer dealers receive worse rates and worse monetary-policy passthrough.
- **Model.** OTC customers bargain with connected dealers using other dealer offers as outside options. More links improve bargaining power but are costly to form.
- **Counterfactuals.** Direct access to inter-dealer platforms or a central-bank secured deposit facility can improve passthrough and reduce rate dispersion.
- **Takeaway.** Monetary policy transmission can be inefficient and unequal even in a fully collateralized, short-term money market because segmentation gives dealers market power.

2 Data and measurement

2.1 Repo market setting

- A repo is a collateralized loan in which one party lends cash against securities collateral.
- In the inter-dealer market, dealer banks trade with each other on centralized platforms.
- In the OTC market, customers such as banks, insurers, pensions, money market funds, mutual funds, hedge funds, and other financial institutions trade bilaterally with dealers.
- The paper studies German, French, Italian, and Spanish government collateral segments.

Interpretation: The contracts look standardized and safe, but the trading structure is segmented. The key friction is not credit risk; it is market access and bargaining power.

2.2 MMSR transaction data

- The ECB Money Market Statistical Reporting dataset records transaction-level repo trades made by large reporting dealers.
- The sample covers February 2017 to February 2020.

- The dataset includes both inter-dealer and OTC trades, collateral information, repo rates, maturity, haircuts, customer sector, and dealer identities.
- This makes it possible to compare dealer rates in the centralized inter-dealer market with rates charged to OTC customers for similar collateral and loan characteristics.

Interpretation: The dataset is unusually powerful because it observes both the dealer-to-dealer benchmark and dealer-to-customer trades in the same market.

2.3 Customer links

- For each customer, the paper counts the number of unique dealers with which the customer trades.
- Most customers have very few dealer relationships.
- The number of links is the paper's main proxy for the customer's bargaining power.

Interpretation: A customer with one dealer has little outside-option value. A customer with several dealers can threaten to trade elsewhere and therefore bargains for a better repo rate.

2.4 Residualized repo rates

To remove mechanical differences in contract characteristics, the paper residualizes repo rates within collateral country and month:

$$r_{it} = X_{it}\beta_t + \varepsilon_{it}. \tag{1}$$

Here X_{it} includes collateral ISIN fixed effects, maturity fixed effects, and haircut controls.

Interpretation: The residualized rate isolates dispersion that remains after controlling for observable loan and collateral characteristics. This dispersion is interpreted as evidence of non-fundamental pricing frictions such as market power.

2.5 Rate dispersion and net interest margins

- Rate dispersion is measured by the volume-weighted standard deviation of repo rates or residualized repo rates.
- Dealer net interest margins are measured as the difference between rates at which dealers lend to OTC customers and rates at which dealers borrow from OTC customers, relative to inter-dealer benchmarks.

Interpretation: Dispersion and margins are central symptoms of dealer market power. In a competitive market, similar contracts should trade at similar rates and passthrough should be close to complete.

3 Empirical specifications

3.1 Customer characteristics and repo rates

The paper estimates residualized rates as a function of the number of dealer links:

$$Rate_{cdm}^{resid} = \beta_1 NumDealers_c + \gamma_m + \delta_s + \omega_d + Maturity_{cdm} + Haircut_{cdm} + \varepsilon_{cdm}. \tag{2}$$

- c indexes customers.
- d indexes dealers.
- m indexes collateral country.

- s indexes customer sector.
- Dealer fixed effects and sector fixed effects control for dealer and customer-sector heterogeneity.

Interpretation: If more links improve bargaining power, the number of dealers should improve customer rates: higher rates when customers lend cash to dealers and lower rates when customers borrow cash from dealers.

3.2 DFR-to-OTC passthrough

The ECB cut the deposit facility rate from -40 bps to -50 bps in September 2019. For each collateral ISIN k, the paper defines DFR-to-OTC passthrough as

$$Passthrough_k^{\mathrm{DFR,OTC}} = \frac{Rate_k^{\mathrm{OTC},post} - Rate_k^{\mathrm{OTC},pre}}{-10}. \tag{3}$$

Interpretation: The denominator is the size of the DFR cut. A passthrough of 100% means the OTC repo rate moves one-for-one with the policy rate cut.

3.3 Inter-dealer to OTC passthrough

The paper then divides DFR-to-OTC passthrough by the DFR-to-ID passthrough:

$$Passthrough_k^{\mathrm{ID,OTC}} = \frac{Passthrough_k^{\mathrm{DFR,OTC}}}{Passthrough_k^{\mathrm{DFR,ID}}}. \tag{4}$$

Interpretation: This isolates the second stage of monetary-policy transmission: how much of the inter-dealer rate movement reaches OTC customers.

3.4 Customer links and passthrough

The customer-level passthrough regression is

$$Passthrough_{cdm}^{\mathrm{ID,OTC}} = \beta_1 NumDealers_c + \gamma_m + \delta_s + \omega_d + Maturity_{cdm} + Haircut_{cdm} + \varepsilon_{cdm}. \tag{5}$$

Interpretation: If market power impairs transmission, customers with more dealer relationships should experience better passthrough.

4 Model and counterfactual framework

4.1 Core model mechanism

- Dealers can trade in the inter-dealer market at rate r_{ID}.
- OTC customers cannot directly access the inter-dealer market and must trade with connected dealers.
- Depositing customers have reservation values for lending cash; borrowing customers have reservation values for borrowing cash.
- Dealers incur balance-sheet costs when intermediating OTC trades.
- Customers bargain with each connected dealer, using offers from other connected dealers as outside options.

Interpretation: More dealer links raise the customer's bargaining power because each dealer must compete with more outside options.

4.2 Bargaining power and number of links

The model estimates the relationship between passthrough and the number of dealer links using

$$Passthrough_c^{IDtoOTC} = \beta_N \mathbf{1}\{NumDealers_c = N\} + \gamma_m + \varepsilon_c. \tag{6}$$

Interpretation: This fitted relationship is used to pin down the bargaining-power parameter in the model.

4.3 Link formation costs

The cost of connecting to the Nth dealer in market segment msp is parameterized as

$$\log\left(C_{msp}^N\right) = \xi_{msp}^1 + \xi_{msp}^2 \log(N-1). \tag{7}$$

Interpretation: The slope parameter captures how quickly link formation becomes more expensive as a customer adds additional dealers.

4.4 Direct platform access

If OTC customers can access inter-dealer platforms at cost C_{ID}, they connect when the platform surplus exceeds the surplus from bargaining with dealers:

$$\Psi_D(r_{ID} - v_D) - C_{ID} \geq \Psi_D((r_{ID} - \tau_D - v_D)(1 - Nt(N))) - \sum_{i=1}^{N} C_i. \tag{8}$$

Interpretation: Free or low-cost platform access gives customers an outside option and can restore more complete passthrough.

4.5 Reverse repo facility

The central bank can offer a secured deposit facility at rate r_{RRP}. If it is below the inter-dealer rate net of dealer costs, it may not be used directly but still improves customers' outside options.

For a depositing customer, the dealer rate becomes

$$r_D(v_D, \theta_D) = \tilde{v}_D + \{(r_{ID} - \tau_D) - \tilde{v}_D\}(1 - Nt(N)), \tag{9}$$

where $\tilde{v}_D = \max(v_D, r_{RRP})$.

Interpretation: A central-bank facility can improve passthrough even without actual take-up because it strengthens customers' bargaining positions.

5 Identification and empirical design

5.1 What identifies market power

- Similar repo contracts trade at different rates after controlling for collateral, maturity, and haircuts.
- Dealers earn positive net interest margins in OTC intermediation.
- Customers with more dealer links receive better rates.
- Policy-rate changes pass through imperfectly from inter-dealer markets to OTC customers.

Interpretation: Each fact is consistent with dealer market power. Together, they make a strong case that market segmentation impedes monetary-policy transmission.

5.2 Why the September 2019 DFR cut is useful

- The DFR cut provides a clear policy-rate shock.
- Short-maturity repos allow the authors to compare rates before and after the cut over a narrow time window.
- The paper compares passthrough in inter-dealer and OTC markets for the same collateral countries.

Interpretation: The event separates the first-stage pass-through from the ECB to the inter-dealer market and the second-stage pass-through from dealers to OTC customers.

5.3 What the paper cannot fully prove

- The model is estimated in a low-rate environment and counterfactuals at high rates require extrapolation.
- Link formation costs and customer valuations are inferred indirectly.
- Unobserved customer characteristics may still affect observed relationships, though residualization and fixed effects reduce this concern.
- The paper focuses on the euro-area repo market and may not directly generalize to all repo markets.

Interpretation: The empirical evidence is strong on segmentation and passthrough; the counterfactuals are model-based quantifications of regulatory alternatives.

6 Tables and figures

Visuals are directly extracted from the paper and grouped compactly to save space.

6.1 Table 1 and Figure 1: market structure and rate dispersion

Read:

- Table 1 shows that most OTC customers have only one or two dealer links across collateral countries and customer sectors.
- Figure 1 shows large dispersion in OTC repo rates and residualized repo rates for German, French, Italian, and Spanish collateral.
- Dispersion declines over time but remains meaningful even after controlling for collateral and loan characteristics.

Economic message: The repo market is segmented: customers have few links and trade at dispersed rates, even for standardized collateralized contracts.

Table 1

OTC repo market structure. This table shows the distribution of trading relationships and repo volumes for OTC repo customers in different sectors, for German (DE), French (FR), Italian (IT) and Spanish (ES) collateral-backed repos. The sectors include insurance and pensions, banks, money market funds, investment funds, other financial institutions, and non-financial institutions. The first three rows of each panel show the first, second, and third quartile of the number of unique dealers for customers in each sector, where the number of unique dealers for each customer is counted from February 2017 to February 2020. The last two rows of each panel show the average monthly volumes that dealers lend and borrow from customers in each sector in €1 billion from February 2017 to February 2020.

a: DE

	Ins./Pens.	Banks	MMF	Funds	Other Fin.	Non-Fin.
No of Links (Q1)	1	1	1	1	1	1
No of Links (Q2)	1	2	1	1	1	1
No of Links (Q3)	3	4	1	2	2	2
Dealer Borrow Volume	22	19	26	202	153	8
Dealer Lend Volume	22	59	16	74	59	53

b: FR

	Ins./Pens.	Banks	MMF	Funds	Other Fin.	Non-Fin.
No of Links (Q1)	1	1	1	1	1	1
No of Links (Q2)	2	2	1	1	1	1
No of Links (Q3)	3	4	1	2	2	2
Dealer Borrow Volume	18	40	28	80	63	27
Dealer Lend Volume	94	58	10	45	31	98

c: IT

	Ins./Pens.	Banks	MMF	Funds	Other Fin.	Non-Fin.
No of Links (Q1)	1	1	1	1	1	1
No of Links (Q2)	1	2	1	1	1	1
No of Links (Q3)	2	3	1	2	2	2
Dealer Borrow Volume	15	36	12	125	36	10
Dealer Lend Volume	18	50	6	166	47	36

d: ES

	Ins./Pens.	Banks	MMF	Funds	Other Fin.	Non-Fin.
No of Links (Q1)	1	1	1	1	1	1
No of Links (Q2)	1	2	1	1	1	1
No of Links (Q3)	1	4	1	1	2	1
Dealer Borrow Volume	36	26	15	186	18	19
Dealer Lend Volume	7	36	9	57	29	16

Table 1: OTC market structure

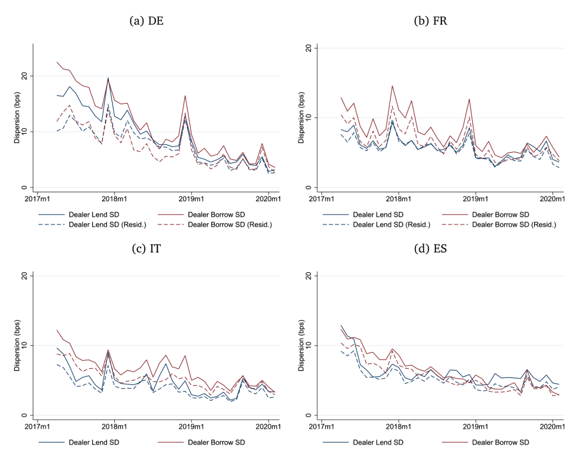


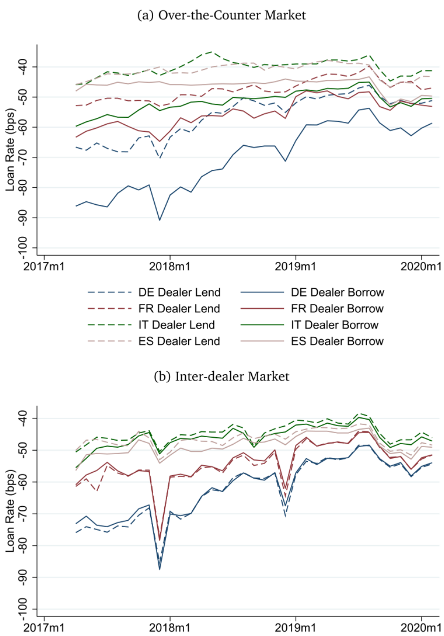
Figure 1: dispersion in OTC rates

6.2 Figure 2, Tables 2–3: dealer rates and customer links

Read:

- Figure 2 compares dealer borrow and lend rates in OTC and inter-dealer markets.
- Tables 2 and 3 show that the number of dealer links improves residualized customer rates.
- Connecting to one more dealer improves dealer-borrow rates by roughly 0.25 bps and reduces dealer-lend rates by roughly 0.49 bps in the most restrictive specifications.

Economic message: Customers with more links have better bargaining power and obtain more favorable repo rates.



(a) Figure 2: OTC and inter-dealer rates.

Table 2

Customer characteristics and residualized repo rates (dealer borrow). This table shows how the residualized repo rates at which dealer borrow from customers vary with the number of links that customers form. For the dependent variable, we first calculate the residualized repo rates for each collateral ISIN. Then, we average the residualized repo rates by collateral country between each dealer-customer pair. Number of Dealers is the number of unique dealers that the customer trades with during our sample period. *FR*, *IT*, and *ES* are dummy variables equal to one if the repo involves French, Italian, and Spanish government collateral, respectively. We control for the average maturity and haircut of repos between each dealer-customer pair. We also include customer-sector and dealer fixed effects and cluster standard errors by dealer. Observations are weighted by one over the number of links that the customer forms.

	(1)	(2)	(3)	(4)
	Residual Borrow Rates			
Number of Dealers	0.227**	0.203**	0.272***	0.249**
	[0.089]	[0.093]	[0.100]	[0.106]
ES	-3.448**	-4.253***	-3.463**	-4.124**
	[1.401]	[1.227]	[1.581]	[1.526]
FR	-0.134	-0.875	0.499	-0.419
	[0.725]	[0.625]	[1.037]	[0.932]
IT	-1.560	-1.662	-1.244	-1.514
	[1.072]	[1.068]	[1.359]	[1.312]
Maturity			-0.011	0.017
			[0.086]	[0.087]
Haircut			-0.360	1.161
			[3.848]	[4.360]
Cntp Sector FE	Yes	Yes	Yes	Yes
Dealer FE	No	Yes	No	Yes
Observations	3536	3531	3105	3098
Adj. R-squared	0.237	0.283	0.250	0.300

(b) Table 2: dealer-borrow rates.

Customer characteristics and residualized repo rates (dealer lend). This table shows how the residualized repo rates at which dealers lend to customers vary with the number of links that customers form. For the dependent variable, we first calculate the residualized repo rates for each collateral ISIN. Then, we average the residualized repo rates by collateral country between each dealer-customer pair. Number of Dealers is the number of unique dealers that the customer trades with during our sample period. *FR*, *IT*, and *ES* are dummy variables equal to one if the repo involves French, Italian, and Spanish government collateral, respectively. We control for the average maturity and haircut of repos between each dealer-customer pair. We also include customer-sector and dealer fixed effects and cluster standard errors by dealer. Observations are weighted by one over the number of links that the customer forms.

	(1)	(2)	(3)	(4)
	Residual Lend Rates			
Number of Dealers	-0.454***	-0.434***	-0.507***	-0.485***
	[0.122]	[0.119]	[0.134]	[0.121]
ES	-2.576	-2.237	-2.145	-1.793
	[1.759]	[1.625]	[1.810]	[1.711]
FR	-2.060**	-1.649**	-2.471***	-2.085***
	[0.826]	[0.668]	[0.851]	[0.651]
IT	-2.724***	-2.953***	-2.620**	-2.637***
	[0.933]	[0.799]	[1.033]	[0.921]
Maturity			0.056**	0.069**
			[0.027]	[0.026]
Haircut			-0.187	-2.390
			[3.024]	[1.918]
Cntp Sector FE	Yes	Yes	Yes	Yes
Dealer FE	No	Yes	No	Yes
Observations	3626	3622	3108	3103
Adj. R-squared	0.056	0.179	0.070	0.232

(c) Table 3: dealer-lend rates.

6.3 Figures 3–4: net interest margins

Read:

- Figure 3 shows dealers' net interest margins for German, French, Italian, and Spanish collateral.
- Figure 4 shows that margins vary by customer sector.
- Money market funds tend to receive worse rates, while banks often receive better rates.

Economic message: Dealers earn meaningful OTC intermediation margins, and rate discrimination differs across customer sectors.

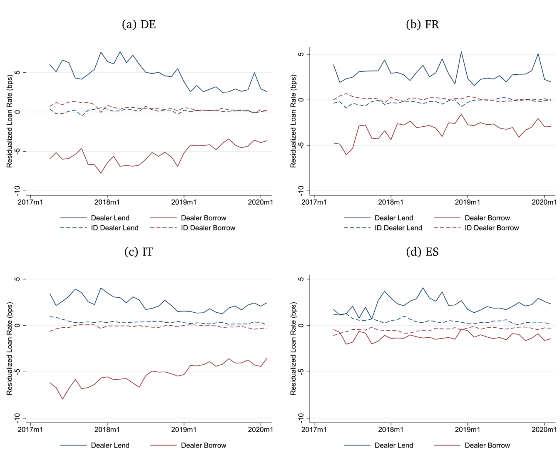


Figure 3: net interest margins

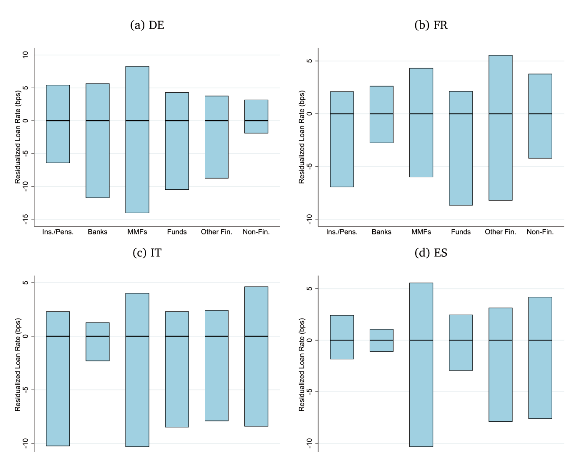


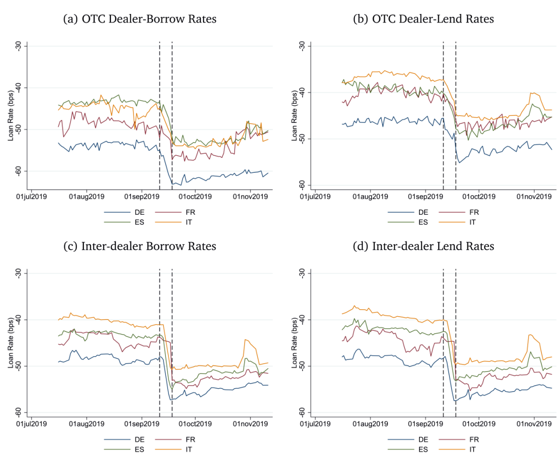
Figure 4: margins by sector

6.4 Figure 5, Table 4, and Table 5: monetary-policy passthrough

Read:

- Figure 5 shows repo rates around the September 2019 ECB rate cut.
- Table 4 shows that ID-to-OTC passthrough is incomplete: about 79.4% for dealer-borrow trades and 71.7% for dealer-lend trades.
- Table 5 shows that customers with more dealer links experience better passthrough.
- The number-of-dealers coefficient is positive and significant in all main passthrough specifications.

Economic message: Segmentation creates inefficient and unequal monetary-policy transmission: customers with fewer links receive worse passthrough.



(a) Figure 5: September 2019 rate cut.

Table 4

Distribution of passthrough. This table shows the distribution of passthroughs from the September 2019 rate cut, expressed in percent. Panel (a) calculates passthrough using repo rates and panel (b) calculates passthrough using residualized repo rates. For each ISIN, the DFR-OTC passthrough is the average change in repo rates in the OTC market from the pre- to post-rate cut period, divided by the deposit facility rate change of -10 bps. The first two rows of each panel show their volume-weighted distribution. For each collateral ISIN, the ID-OTC passthrough is the average change in repo rates in the OTC market from the pre- to post-rate cut period, divided by that in the inter-dealer market from the pre- to post-rate cut period. The last two rows of each panel show the volume-weighted distribution of ID-OTC passthroughs.

a: Repo Rates

	Average	p10	p25	p50	p75	p90
DFR to OTC, Dealer Lend	59.7	24.4	38.1	69.1	78.2	84.7
DFR to OTC, Dealer Borrow	70.3	18.6	61.5	79.8	85.8	94.9
ID to OTC, Dealer Lend	71.7	29.4	49.9	73.4	89	101.1
ID to OTC, Dealer Borrow	79.4	21.8	67.1	83.9	100.3	115.5

b: Residualized Repo Rates

	Average	p10	p25	p50	p75	p90
DFR to OTC, Dealer Lend	58.4	20.9	36.2	67.1	76.3	85.9
DFR to OTC, Dealer Borrow	69.6	17.4	62.3	77.6	89.6	98.4
ID to OTC, Dealer Lend	72.1	27.5	41.1	77.6	94.4	101.7
ID to OTC, Dealer Borrow	79	19.7	63.4	83.8	99.2	108.5

(b) Table 4: passthrough distribution.

Table 5

Customer characteristics and passthrough. This table shows how ID-OTC passthrough to customers varies with the number of links that customers form. For the dependent variable, we first calculate ID-OTC passthrough for each collateral ISIN as the average change in residualized repo rates in the OTC market from the pre- to post-rate cut period, divided by that in the inter-dealer market from the pre- to post-rate cut period. Then, we average ID-OTC passthrough by collateral country between each dealer-customer pair. Number of Dealers is the number of unique dealers that the customer trades with during the pre rate-cut month. *FR*, *IT*, and *ES* are dummy variables equal to one if the repo involves French, Italian, and Spanish government collateral, respectively. We control for the average maturity and haircut of repos between each dealer-customer pair. We also include customer-sector and dealer fixed effects and cluster standard errors by dealer. Observations are weighted by one over the number of links that the customer forms.

	(1) Passthrough	(2)	(3)	(4)
Number of Dealers	2.788*	3.893***	2.673*	3.958***
	[1.510]	[1.230]	[1.493]	[1.212]
ES	-41.920**	-43.757**	-38.137*	-39.553*
	[18.194]	[18.969]	[18.741]	[19.172]
FR	6.612	5.700	8.962	8.220
	[12.458]	[13.333]	[12.187]	[12.738]
IT	2.358	-3.779	4.937	-2.303
	[7.090]	[5.479]	[6.882]	[4.924]
Dealer Lend	-5.310	-1.785	-3.936	-0.812
	[3.398]	[3.468]	[3.340]	[3.333]
Haircut			-33.977	4.801
			[31.779]	[17.987]
Maturity			-0.685	-0.533
			[0.465]	[0.392]
Cntp Sector	Yes	Yes	Yes	Yes
Dealer FE	No	Yes	No	Yes
Observations	808	804	805	801
Adj. R-squared	0.231	0.354	0.233	0.356

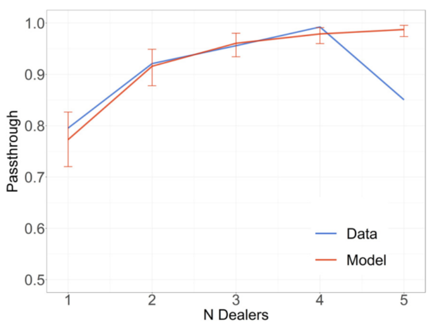
6.5 Figure 6, Table 6, Figure 7, and Table 7: model estimation and fit

Read:

- Figure 6 shows that the model matches the relationship between passthrough and the number of dealer links.

- Table 6 reports estimated bargaining power, customer-value dispersion, balance-sheet costs, and link-formation costs.
- Figure 7 shows that the model fits the relation between trading volume and number of links across sectors.
- Table 7 shows that the model matches net interest margins and rate dispersion reasonably well.

Economic message: The model rationalizes the empirical facts by combining dealer bargaining power, customer heterogeneity, and endogenous link formation.



5. Model fit: relationship between passthrough and the number of linl
figure shows the passthrough from the inter-dealer repo rate to the O'
rate for depositing customers with different number of links. The blue li

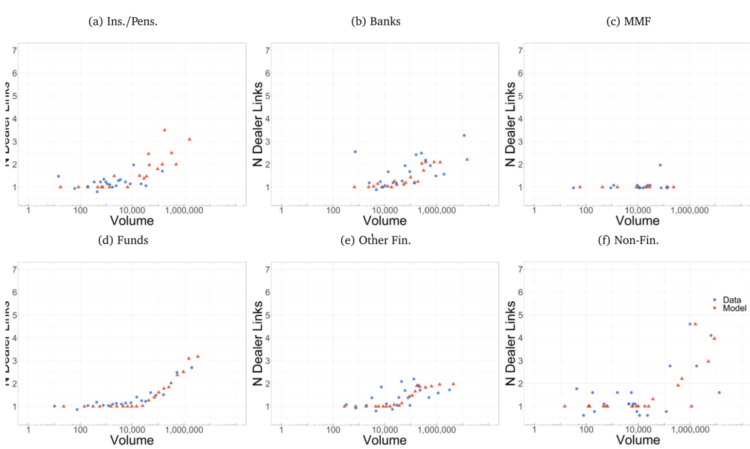
(a) Figure 6: passthrough and links.

Table 6

Parameter estimates. This table shows the parameter estimates of our model for repo depositors. ϕ is the bargaining power of dealers over customers in each bilateral negotiation. μ and σ are the mean and standard deviation of customers' values, respectively, in basis points. τ is the weighted average balance sheet cost. $exp(\zeta^1)$ and ζ^2 are the connection cost parameters in (15). We report $exp(\zeta^1)$, the cost of connecting to the second dealer, in units of €1K per year. We report values as NA when $exp(\zeta^1)$ is above €1 billion annually, or when ζ^2 is above 1000. We estimate $\mu, \sigma, \tau, \zeta^1, \zeta^2$ separately by customer sector and collateral country, but we report averages of parameters across collateral countries, weighting collateral countries within a customer sector by the number of customers in the segment. Parentheses show 95% confidence intervals from a nonparametric bootstrap with 100 replications.

Sector	ϕ	μ	σ	τ	$exp(\zeta^1)$	ζ^2
Banks	0.23	-48.1	19.6	0.31	42.0	3.8
	(0.17, 0.28)	(-61.5, -41.9)	(10.0, 36.8)	(0.02, 0.64)	(5.3, 69.4)	(0.1, 64.0)
Funds		-46.2	20.0	0.37	17.7	1.9
		(-53.2, -41.8)	(14.1, 30.0)	(0.08, 0.55)	(9.8, 42.7)	(0.2, 25.4)
Ins./Pens.		-46.3	4.9	0.07	NA	NA
		(-49.3, -40.4)	(5.1, 15.8)	(0.00, 0.69)	(NA, NA)	(NA, NA)
MMF		-42.6	10.6	0.04	228.8	NA
		(-44.3, -40.0)	(3.6, 21.6)	(0.00, 3.02)	(7.1, NA)	(NA, NA)
Non-Fin.		-46.6	11.9	0.23	9.0	0.1
		(-52.2, -41.0)	(9.9, 37.5)	(0.01, 1.86)	(2.3, 43.4)	(0.0, 36.2)
Other Fin.		-47.3	34.9	0.16	36.8	4.4
		(-54.8, -41.1)	(23.9, 56.7)	(0.05, 1.57)	(4.6, 118.6)	(1.7, 61.4)

(b) Table 6: parameter estimates.



g. 7. Model fit: relationship between the number of links and trading volume. This figure shows binscatter plots of trading volume versus the number of dealer
ıks formed for repo depositors in different sectors in the pre-rate-cut month. The blue and red dots correspond to the data and model predictions, respectively.
the data, flow repo volumes are converted to average outstanding volumes by multiplying trade size by the maturity in days, and then dividing by the number

(a) Figure 7: trading volume and links.

Model fit: net interest margin and dispersion. This table shows the net interest margin and dispersion of repo rates for each sector for repo depositors. Sector is the sector of the customer. Target SD is the observed dispersion in residual repo rates in the data. Model SD is the dispersion in residual repo rates predicted by the model. Target NIM is the average net interest margin of residualized repo rates in the data. Model NIM is the average net interest margin in the model. All quantities are in bps. We present results for different sectors averaged across collateral country, where the weights are the number of customers in that segment in the data. Parentheses show 95% confidence intervals from a nonparametric bootstrap with 100 replications.

Sector	Target SD	Model SD	Target NIM	Model NIM
Banks	4.48	2.49	2.49	3.74
		(1.97, 3.40)		(2.98, 4.36)
Funds	3.64	2.71	2.69	3.53
		(2.36, 3.15)		(3.23, 3.91)
Ins./Pens.	2.28	0.87	0.29	1.47
		(0.82, 1.99)		(1.16, 3.63)
MMF	2.28	1.54	1.62	2.17
		(0.55, 2.74)		(0.75, 5.41)
Non-Fin.	2.86	1.92	1.52	2.59
		(1.62, 3.00)		(2.23, 4.74)
Other Fin.	5.58	4.14	3.97	5.24
		(3.17, 5.17)		(4.32, 6.73)

ɾce of error is that the fitted model produces dispersions sligh
er than those in the data, and net interest margins slightly lar

(b) Table 7: model fit for margins and dispersion.

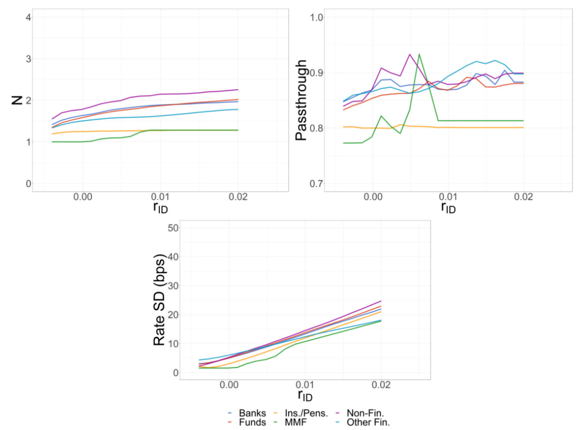
6.6 Figure 8, Table 8, and Table 9: counterfactual policies

Read:

- Figure 8 shows that higher inter-dealer rates lead to only modest increases in link formation and only modest improvements in passthrough.

- Table 8 shows that direct inter-dealer platform access improves passthrough when access costs are low; free access yields 100% passthrough and eliminates rate dispersion.
- Table 9 shows that a reverse repo facility improves passthrough and reduces dispersion even when it is mainly used as an outside option.

Economic message: Regulatory access tools can offset dealer market power by giving customers better outside options.



ks and passthrough with rate increases. This figure shows the number of dealer links formed, the inter-dealer to OTC pass
iodel predicts as monetary policy increases the inter-dealer rate. Each line corresponds to the results for a different sector. '
veraged across collateral country, where the weights are the number of customers in that segment in the data.

(a) Figure 8: rate increases and passthrough.

Table 8

Effect of inter-dealer platform access. This table shows results from the inter-dealer platform access counterfactual. Baseline refers to the benchmark case without any inter-dealer platform access. Access cost 100K, 25K, 5K, and 0K refer to the counterfactual cases where inter-dealer platform access is available at an annual cost of €100K, €25K, €5K, and €0K, respectively. Rate disp is the standard deviation of rates, in bps. Passthrough is the derivative of average repo rates with respect to inter-dealer repo rates. Frac. Connected is the fraction of agents who optimally connect to the inter-dealer platform. Parentheses show 95% confidence intervals from a nonparametric bootstrap with 100 replications.

Type	Rate Disp	Passthrough	Frac. Connected
Baseline	3.01	82.9%	
	(2.88, 3.52)	(78.6%, 87.4%)	
100k	3.13	84.5%	12.8%
	(2.99, 3.56)	(81.1%, 88.3%)	(10.8%, 15.3%)
25k	3.10	86.3%	27.0%
	(2.94, 3.51)	(83.7%, 89.7%)	(24.1%, 28.6%)
5k	2.62	90.4%	42.6%
	(2.52, 3.14)	(88.6%, 92.8%)	(39.9%, 44.9%)
0	0.00	100.0%	100.0%
	(0.00, 0.00)	(100.0%, 100.0%)	(100.0%, 100.0%)

(b) Table 8: platform access.

Table 9

Effect of the reverse repurchase facility. This table shows results from the RRP counterfactual on the inter-dealer to OTC passthrough and rate dispersion. Baseline refers to the benchmark case without the RRP. 50 bps, 25 bps, and 10 bps refer to the counterfactual cases where Δ_{RRP}, the gap between the RRP rate and the inter-dealer rate minus balance sheet costs, is 50 bps, 25 bps, and 10 bps, respectively. Rate disp is the standard deviation of rates, in bps. Passthrough is the derivative of average repo rates with respect to inter-dealer repo rates. Frac. Binding is the fraction of agents for whom the RRP is a binding outside option. Parentheses show 95% confidence intervals from a nonparametric bootstrap with 100 replications.

Type	Rate Disp.	Passthrough	Frac. Binding
Baseline	3.01	82.9%	
	(2.88, 3.52)	(78.6%, 87.4%)	
50 bps	2.75	83.7%	5.2%
	(2.57, 3.04)	(79.2%, 88.7%)	(2.0%, 12.2%)
25 bps	2.01	88.2%	25.1%
	(1.65, 2.33)	(84.1%, 92.6%)	(17.6%, 40.1%)
10 bps	0.92	96.5%	59.6%
	(0.73, 1.46)	(94.9%, 98.0%)	(52.4%, 70.6%)

(c) Table 9: RRP facility.

7 Short summary

- Repo markets are an important first stage of monetary-policy transmission.
- The euro-area OTC repo market is segmented because non-dealer customers rely on dealer intermediation.
- Customers typically have few dealer links, giving dealers market power.
- Similar repo contracts trade at dispersed rates even after residualizing for collateral and loan characteristics.
- Dealers earn positive net interest margins in OTC repo intermediation.
- Customers with more dealer links obtain better borrowing and lending rates.
- The September 2019 ECB rate cut passes through incompletely from inter-dealer markets to OTC customers.
- Passthrough is worse for customers with fewer dealer relationships.
- The model explains these facts with bilateral bargaining and endogenous link formation.
- Direct inter-dealer platform access and a central-bank secured deposit facility can improve passthrough and reduce rate dispersion.

8 Conclusion

8.1 Core conclusion

Eisenschmidt, Ma, and Zhang show that monetary-policy transmission in repo markets is limited by market segmentation and dealer market power. Repo contracts are collateralized and standardized, but OTC customers lack direct market access and depend on a few dealer relationships. As a result, policy-rate changes pass through incompletely and unequally.

8.2 Economic intuition

The intuition is that dealers act as gatekeepers between centralized inter-dealer markets and OTC customers. A customer with few dealer links has weak bargaining power, so a dealer can keep part of the surplus and transmit only part of a policy-rate change. More links improve bargaining power, but links are costly to form, so customers do not generally eliminate dealer market power.

8.3 Takeaway

The paper highlights a liability-side and market-structure channel in monetary-policy transmission. The central bank's policy rate may move inter-dealer rates quickly, but customers' actual funding and deposit rates depend on market access, bargaining power, and available outside options.